

Karnaugh Map

Karnaugh Map

The Karnaugh map, also known as the K-map, is a method to simplify Boolean algebra expressions.

The Karnaugh map reduces the need for extensive calculations by taking advantage of humans' pattern-recognition capability.

Karnaugh Map: A graphical technique for simplifying a Boolean expression into either form:

- minimal sum of products (MSP)
- minimal product of sums (MPS)

Goal of the simplification.

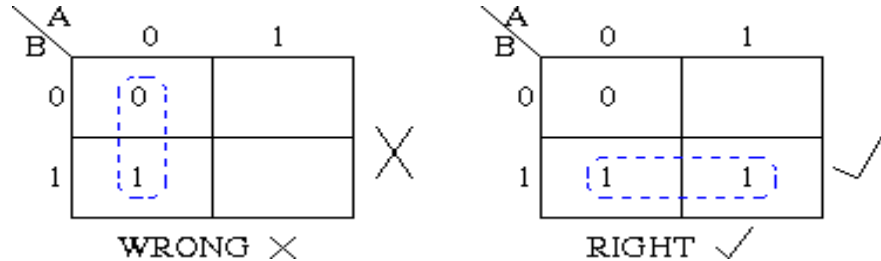
- There are a minimal number of product/sum terms
- Each term has a minimal number of literals

These terms can be used to write a minimal Boolean expression representing the required logic.

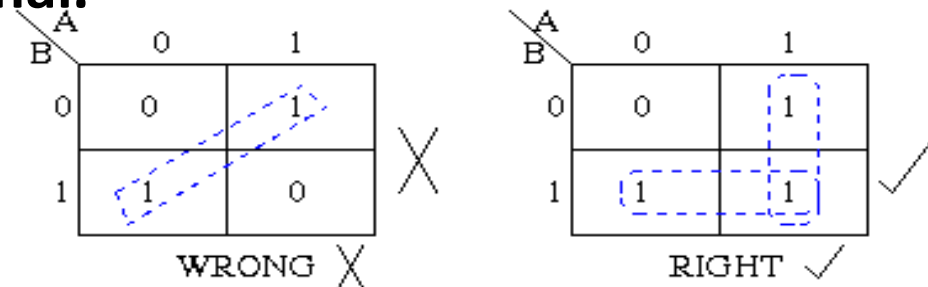
Karnaugh Maps - Rules of Simplification

The Karnaugh map uses the following rules for the simplification of expressions by grouping together adjacent cells containing *ones*.

1.Groups may not include any cell containing a zero.



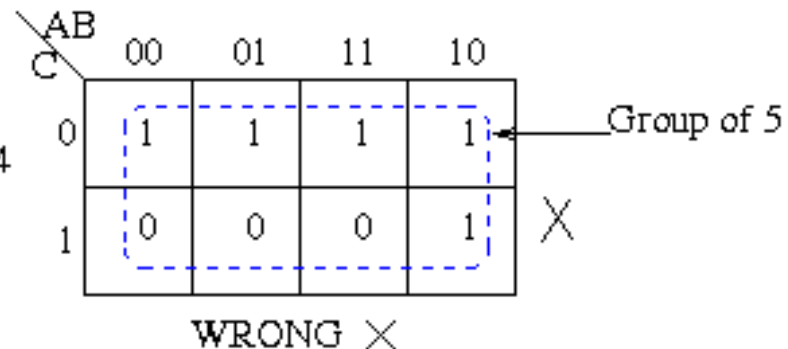
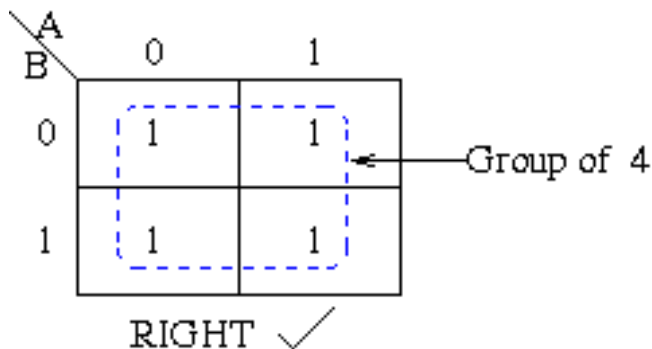
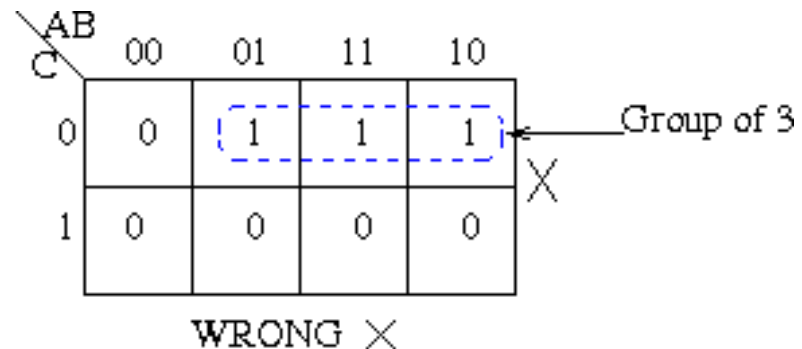
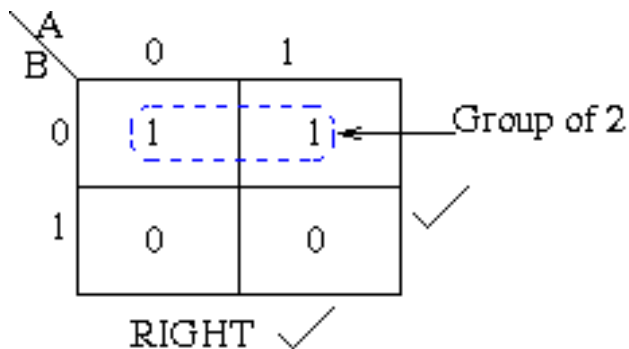
2.Groups may be horizontal or vertical, but not diagonal.



3. Groups must contain 1, 2, 4, 8 cells or in general 2^n cells.

That is if $n = 1$, a group will contain two 1's since $2^1 = 2$.

If $n = 2$, a group will contain four 1's since $2^2 = 4$.



4. Each group should be as large as possible.

$\overline{C}$ \ AB	00	01	11	10
0	1	1	1	1
1	0	0	1	1

RIGHT ✓

$\overline{C}$ \ AB	00	01	11	10
0	1	1	1	1
1	0	0	1	1

WRONG ✗

(Note that no Boolean laws broken,
but not sufficiently minimal)

5. Each cell containing a one must be in at least one group.

$\overline{C}$ \ AB	00	01	11	10
0	0	0	1	1
1	0	0	0	1

Group I

Group II

1 present in at least one group.

6.Groups may overlap.

AB		00	01	11	10
C					
0		1	1	1	1
1		0	0	1	1

Groups overlapping. ✓

RIGHT ✓

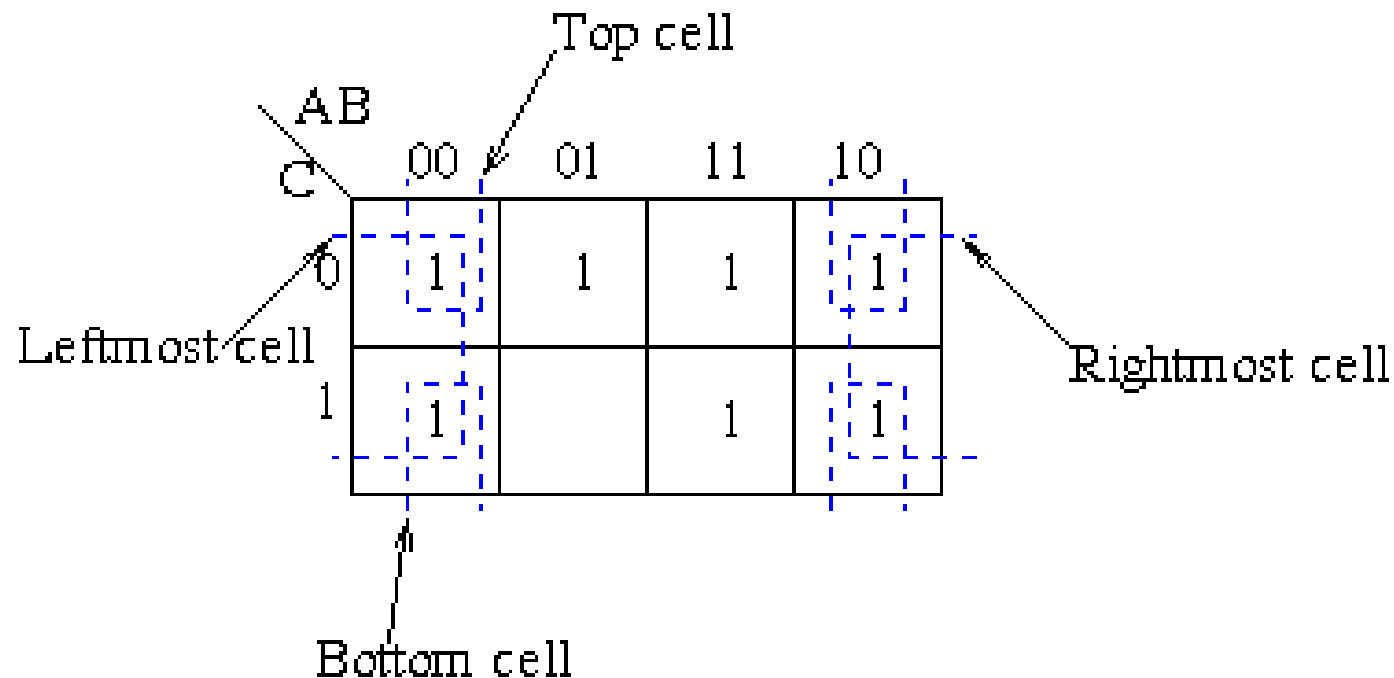
AB		00	01	11	10
C					
0		1	1	1	1
1		0	0	1	1

Groups not overlapping. ✗

WRONG ✗

7.Groups may wrap around the table.

The leftmost cell in a row may be grouped with the rightmost cell and the top cell in a column may be grouped with the bottom cell.



8. There should be as few groups as possible, as long as this does not contradict any of the previous rules.

C \ AB	AB			
	00	01	11	10
0	1	1	1	1
1	0	0	1	1

RIGHT

C \ AB	AB			
	00	01	11	10
0	1	1	1	1
1	0	0	1	1

WRONG

Summmmary:

- No zeros allowed.
- No diagonals.
- Only power of 2 number of cells in each group.
- Groups should be as large as possible.
- Every one must be in at least one group.
- Overlapping allowed.
- Wrap around allowed.
- Fewest number of groups possible.

Examples

Two-Variable K-Map

		a	
		0	1
b	0		
	1		

Karnaugh maps

- Karnaugh maps, or K-maps, are often used to simplify logic problems with 2, 3 or 4 variables.

Cell = 2^n ,where n is a number of variables

For the case of 2 variables, we form a map consisting of $2^2=4$ cells as shown in Figure

		A	
		0	1
B	0	$A + B$	$\bar{A} + B$
	1	$A + \bar{B}$	$\bar{A} + \bar{B}$

Maxterm

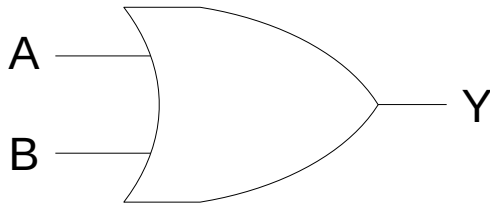
		A	
		0	1
B	0	00 0	10 2
	1	01 1	11 3

		A	
		0	1
B	0	$\bar{A}\bar{B}$	$A\bar{B}$
	1	$\bar{A}B$	AB

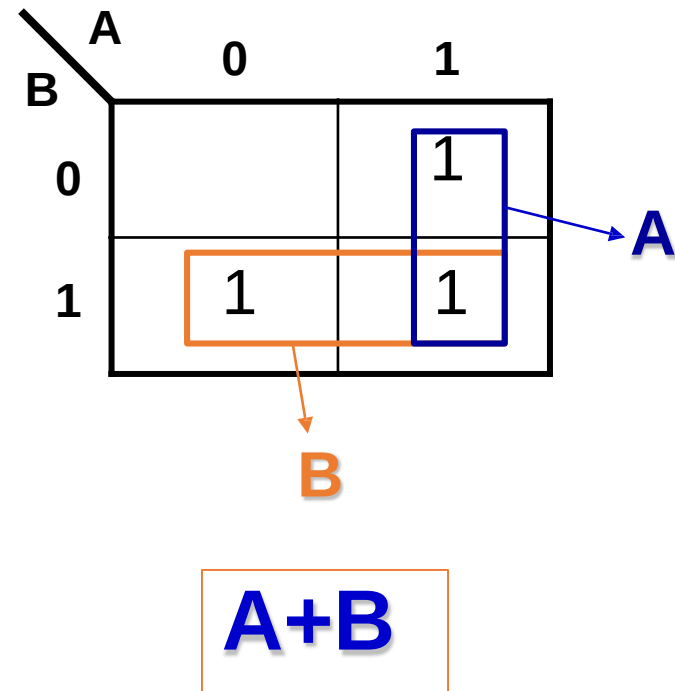
Minterm

Example

2-variable Karnaugh maps are trivial but can be used to introduce the methods you need to learn. The map for a 2-input OR gate looks like this:



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



Example 1:

Consider the following map. The function plotted is: $Z = f(A,B) = AB' + AB$

B \ A	0	1
	0	1
0		1
1		1

Note that values of the input variables form the rows and columns.

That is the logic values of the variables A and B (with one denoting true form and zero denoting false form) form the head of the rows and columns respectively.

The map displayed is a one dimensional type which can be used to simplify an expression in two variables.

There is a two-dimensional map that can be used for up to four variables, and a three-dimensional map for up to six variables.

Referring to the map above, the two adjacent 1's are grouped together.

Through inspection it can be seen that variable B has its true and false form within the group.

This eliminates variable B leaving only variable A which only has its true form.

The minimized answer therefore is $Z = A$.

Karnaugh maps

- 3 variables Karnaugh map

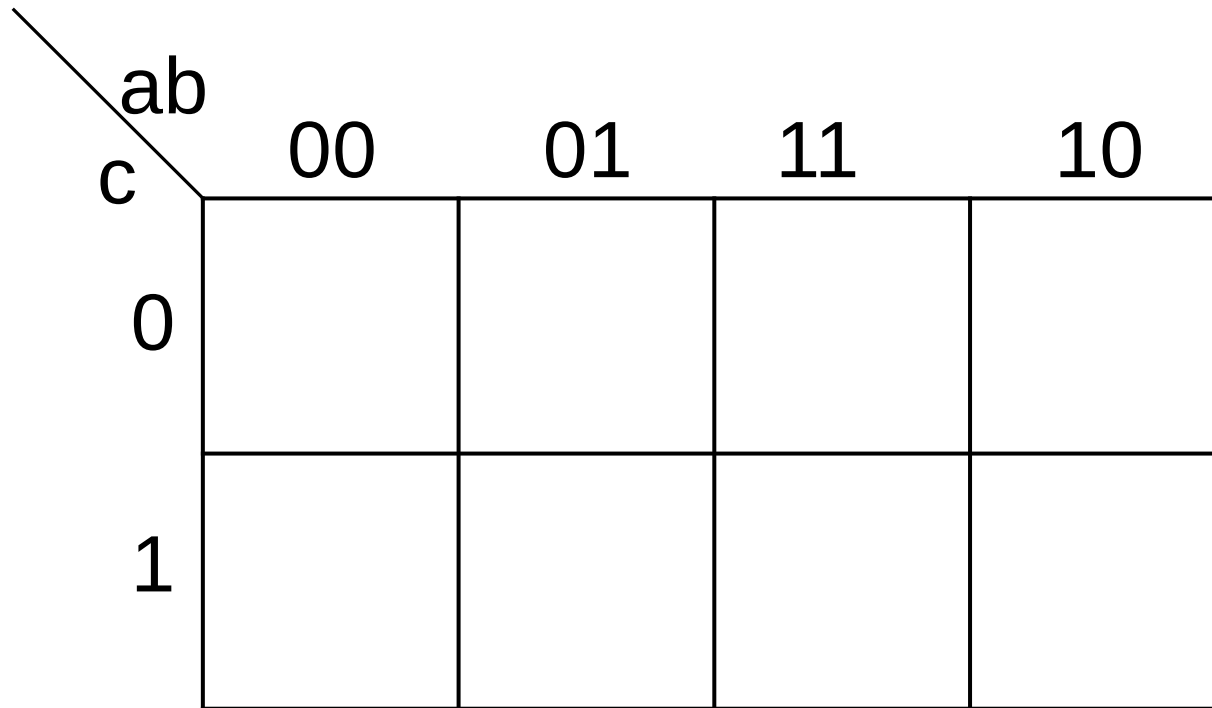
$$\text{Cell} = 2^3 = 8$$

$\begin{array}{c} \text{AB} \\ \diagdown \\ \text{C} \end{array}$		AB			
		00	01	11	10
C	0	$\bar{A}\bar{B}\bar{C}$	$\bar{A}B\bar{C}$	$AB\bar{C}$	$A\bar{B}\bar{C}$
	1	$\bar{A}\bar{B}C$	$\bar{A}BC$	ABC	$A\bar{B}C$

Three-Variable K-Map

		ab			
		00	01	11	10
c	0	m_0	m_2	m_6	m_4
	1	m_1	m_3	m_7	m_5

Three-Variable K-Map



ab \ c	00	01	11	10
0				
1				

Edges are adjacent



Terminology

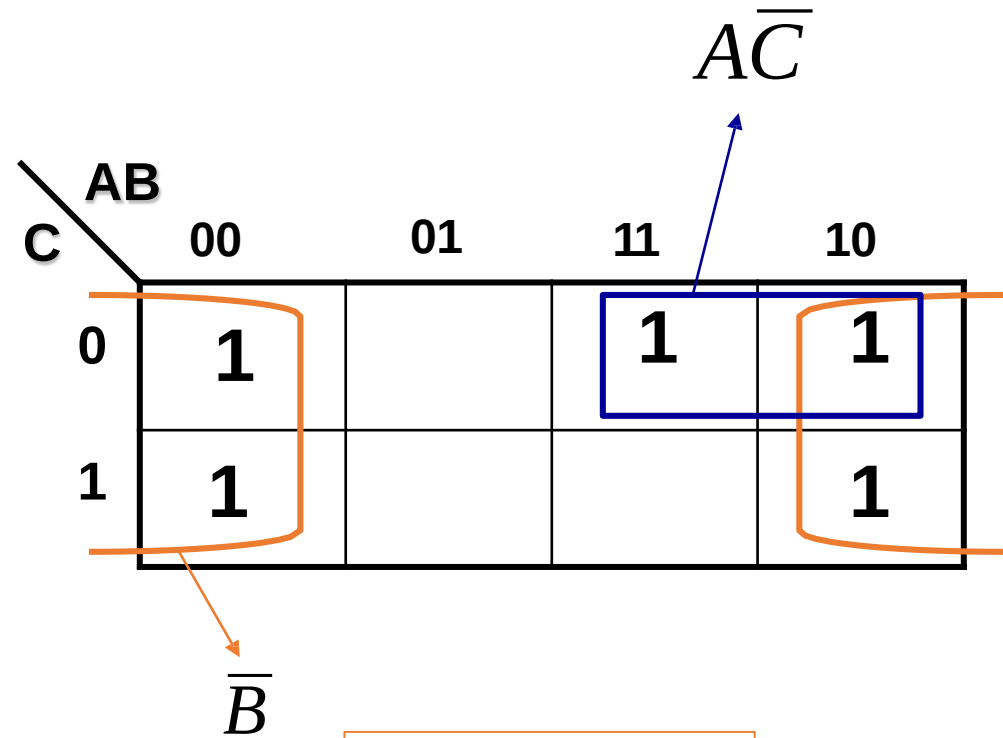
- Implicant
 - Product term that could be used to cover minterms of a function
- Prime Implicant
 - An implicant that is not part of another implicant
- Essential Prime Implicant
 - An implicant that covers at least one minterm that is not contained in another prime implicant

Guidelines for Simplifying Functions

- Group as many squares as possible. This eliminates the most variables.
- Make as few groups as possible. Each group represents a separate product term.
- You must **cover** each minterm at least once. However, it may be covered more than once.

Example

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



$$\overline{B} + A\overline{C}$$

Three-Variable K-Map Example

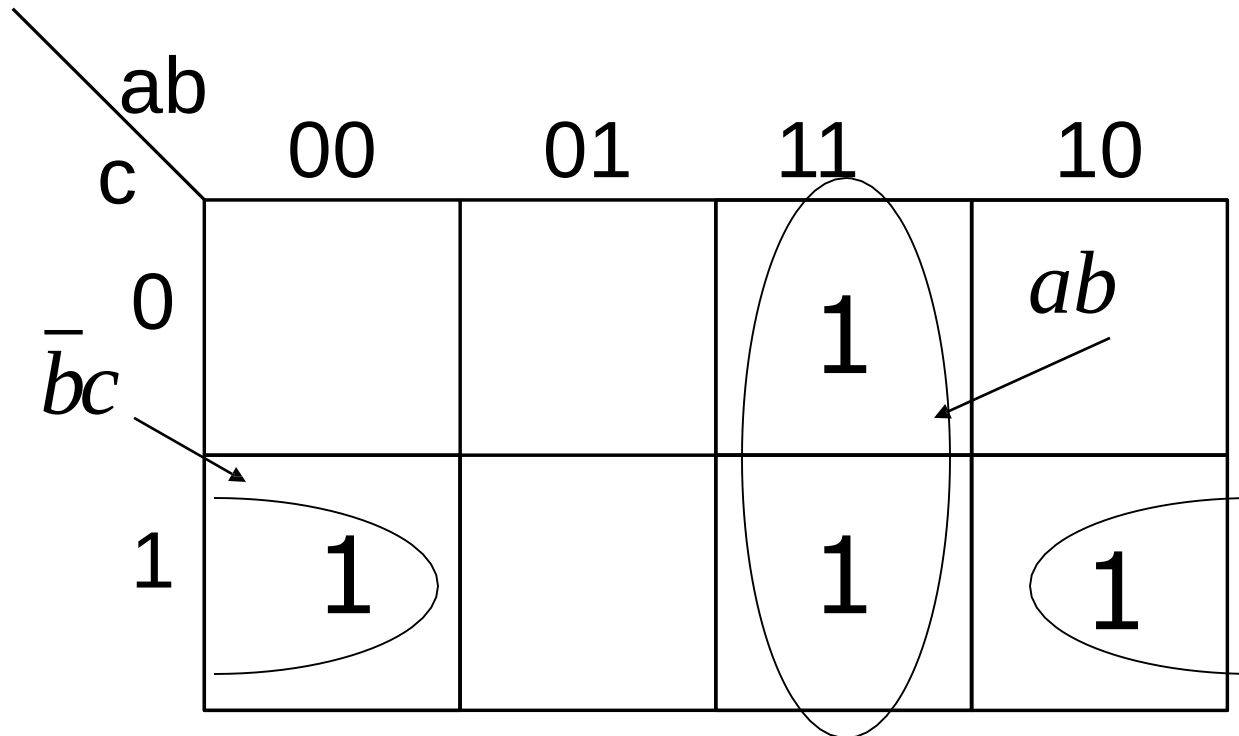
Plot 1's (minterms) of switching function

ab \ c		ab			
		00	01	11	10
c	0			1	
	1	1	1		1

$$F(a,b,c) = \sum m(1,3,5,6)$$

Three-Variable K-Map Example

Plot 1's (minterms) of switching function



$$F(a, b, c) = ab + \bar{b}c$$

Three-Variable K-Maps

$$f = \sum (0,4) = \overline{B}\overline{C}$$

		BC			
		00	01	11	10
A	0	1	0	0	0
	1	1	0	0	0

$$f = \sum (4,5) = A\overline{B}$$

		BC			
		00	01	11	10
A	0	0	0	0	0
	1	1	1	0	0

$$f = \sum (0,1,4,5) = \overline{B}$$

		BC			
		00	01	11	10
A	0	1	1	0	0
	1	1	1	0	0

$$f = \sum (0,1,2,3) = \overline{A}$$

		BC			
		00	01	11	10
A	0	1	1	1	1
	1	0	0	0	0

$$f = \sum (0,4) = \overline{A}C$$

		BC			
		00	01	11	10
A	0	0	1	1	0
	1	0	0	0	0

$$f = \sum (4,6) = A\overline{C}$$

		BC			
		00	01	11	10
A	0	0	0	0	0
	1	1	0	0	1

$$f = \sum (0,2) = \overline{A}\overline{C}$$

		BC			
		00	01	11	10
A	0	1	0	0	1
	1	0	0	0	0

$$f = \sum (0,2,4,6) = \overline{C}$$

		BC			
		00	01	11	10
A	0	1	0	0	1
	1	1	0	0	1

Three-Variable K-Map Examples

A \ BC	BC			
	00	01	11	10
0		1		
1	1		1	1

A \ BC	BC			
	00	01	11	10
0	1		1	1
1	1			1

A \ BC	BC			
	00	01	11	10
0			1	1
1	1	1		

A \ BC	BC			
	00	01	11	10
0			1	
1	1		1	1

A \ BC	BC			
	00	01	11	10
0		1	1	1
1		1	1	

A \ BC	BC			
	00	01	11	10
0	1		1	
1		1	1	1

Example

- Use a K-Map to simplify the following Boolean expression

$$F(a, b, c) = \sum m(2, 3, 6, 7)$$

Three-Variable K-Map Example

Step 1: Plot the K-map

ab \ c		ab			
		00	01	11	10
c	0		1	1	
	1		1	1	

$$F(a,b,c) = \sum m(2,4,5,7)$$

Three-Variable K-Map Example

Step 2: Circle Prime Implicants

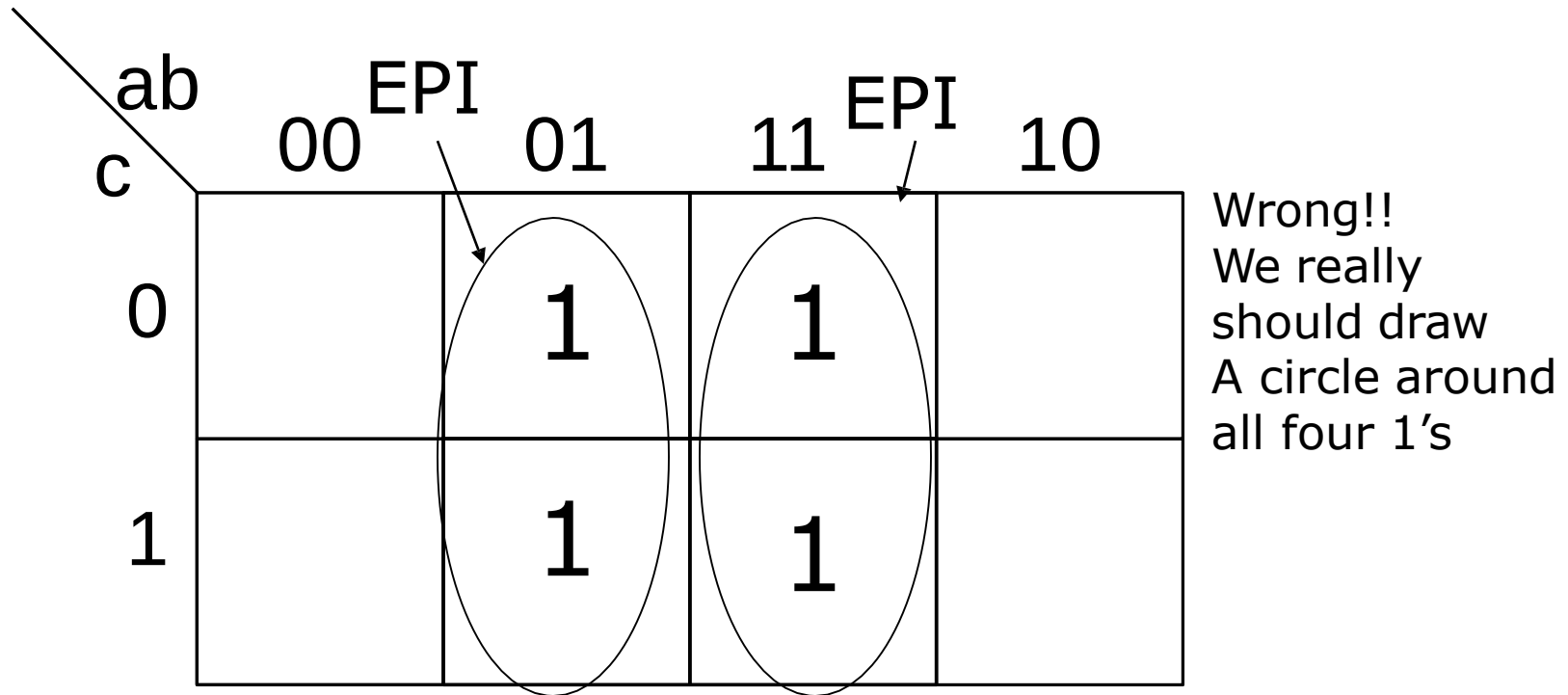
ab \ c		ab			
		00	01	11	10
c	0		1	1	
	1		1	1	

Wrong!!
We really
should draw
A circle around
all four 1's

$$F(a,b,c) = \sum m(2,3,6,7)$$

Three-Variable K-Map Example

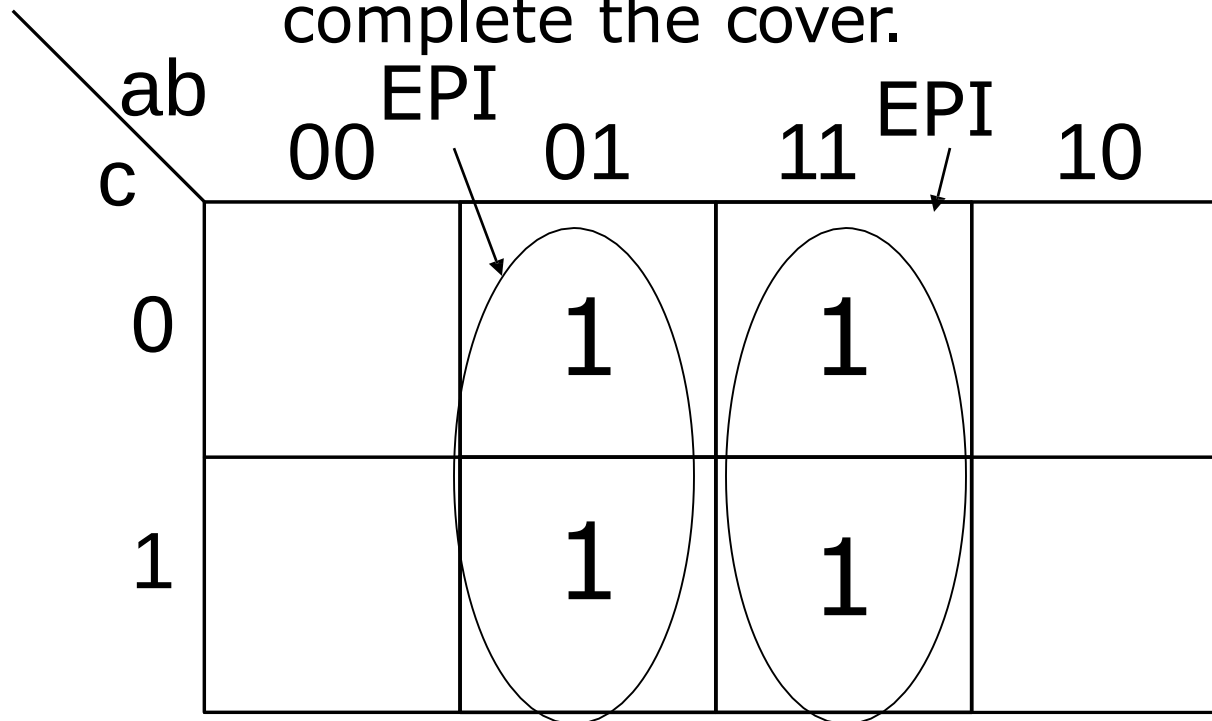
Step 3: Identify Essential Prime Implicants



$$F(a,b,c) = \sum m(2,3,6,7)$$

Three-Variable K-Map Example

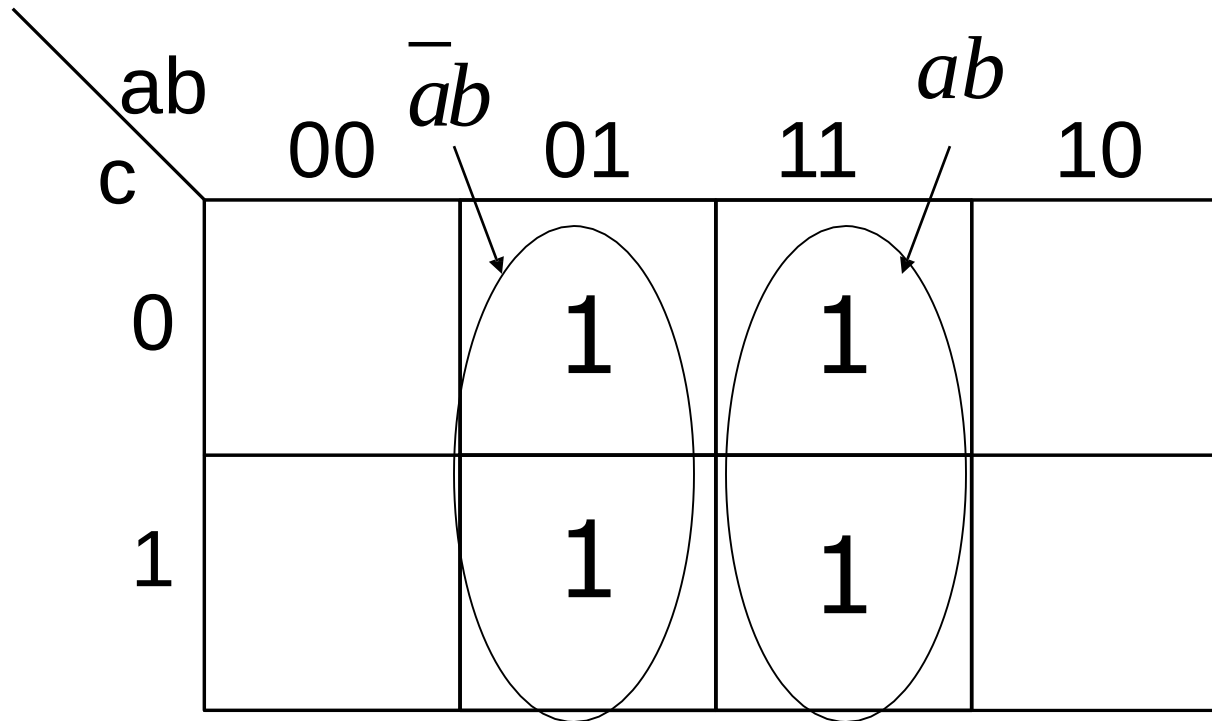
Step 4: Select Remaining Prime Implicants to complete the cover.



$$F(a,b,c) = \sum m(2,3,6,7)$$

Three-Variable K-Map Example

Step 5: Read the map.

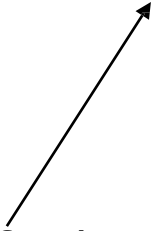


$$F(a,b,c) = \sum m(2,3,6,7)$$

Solution

$$F(a, b, c) = \bar{a}b + ab = b$$

Since we can still simplify the function
this means we did not use the largest
possible groupings.



$$F(a, b, c) = b$$

		<i>ab</i>			
		00	01	11	10
<i>c</i>	0		1	1	
	1		1	1	

b

$$F(a, b, c) = \sum m(2, 3, 6, 7)$$

Special Cases

Three-Variable K-Map Example

		ab			
		00	01	11	10
c	0	1	1	1	1
	1	1	1	1	1

$$F(a,b,c)=1$$

Three-Variable K-Map Example

		ab			
		00	01	11	10
c	0	0	0	0	0
	1	0	0	0	0

$$F(a, b, c) = 0$$

Three-Variable K-Map Example

		ab			
		00	01	11	10
c	0		1		1
	1	1		1	

$$F(a,b,c) = a \oplus b \oplus c$$

Example

- Use a K-Map to simplify the following Boolean expression

$$F(a, b, c) = \sum m(1, 2, 3, 5, 6)$$

Three-Variable K-Map Example

Step 1: Plot the K-map

ab \ c		ab			
		00	01	11	10
c	0		1	1	
	1	1	1		1

$$F(a,b,c) = \sum m(1,2,3,5,6)$$

Three-Variable K-Map Example

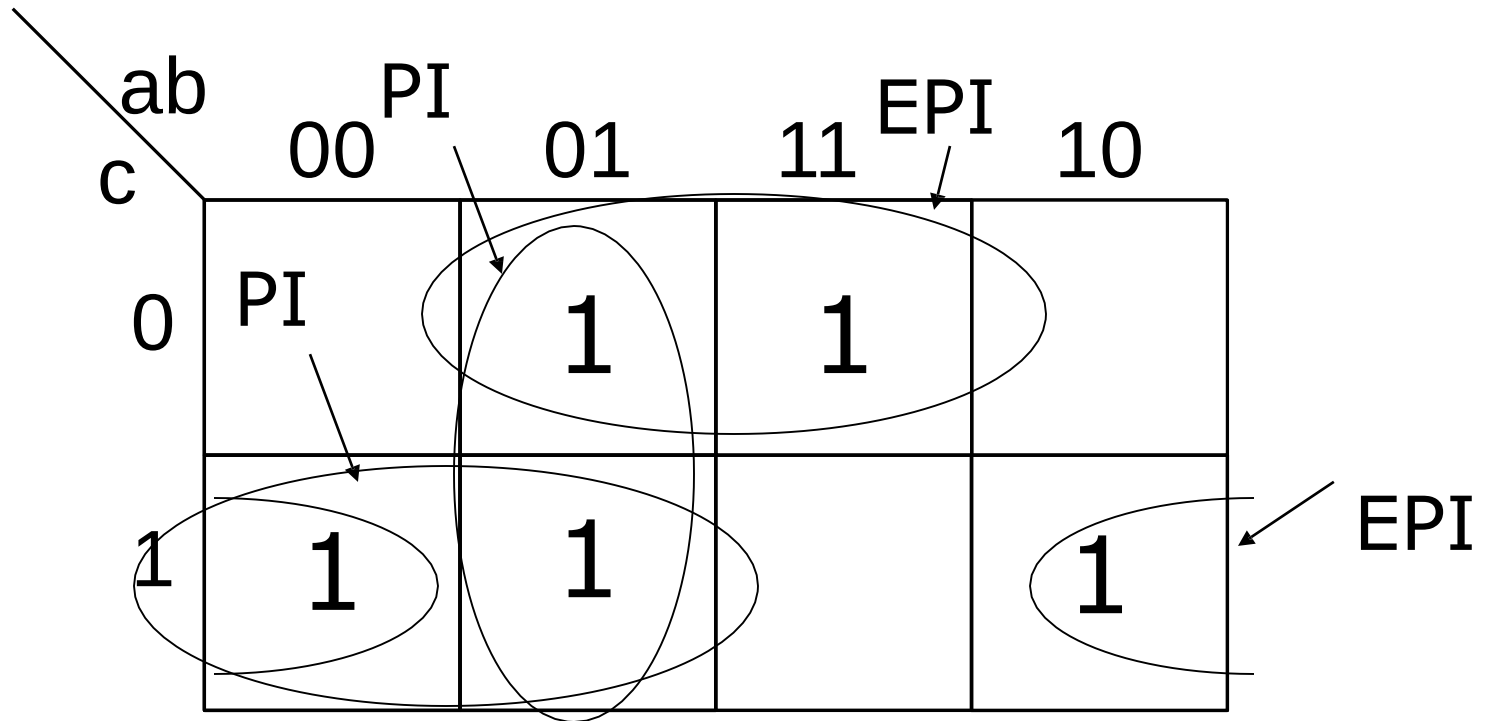
Step 2: Circle **ALL** Prime Implicants

ab \ c		ab			
		00	01	11	10
c	0		1	1	
	1	1	1		1

$$F(a,b,c) = \sum m(1,2,3,5,6)$$

Three-Variable K-Map Example

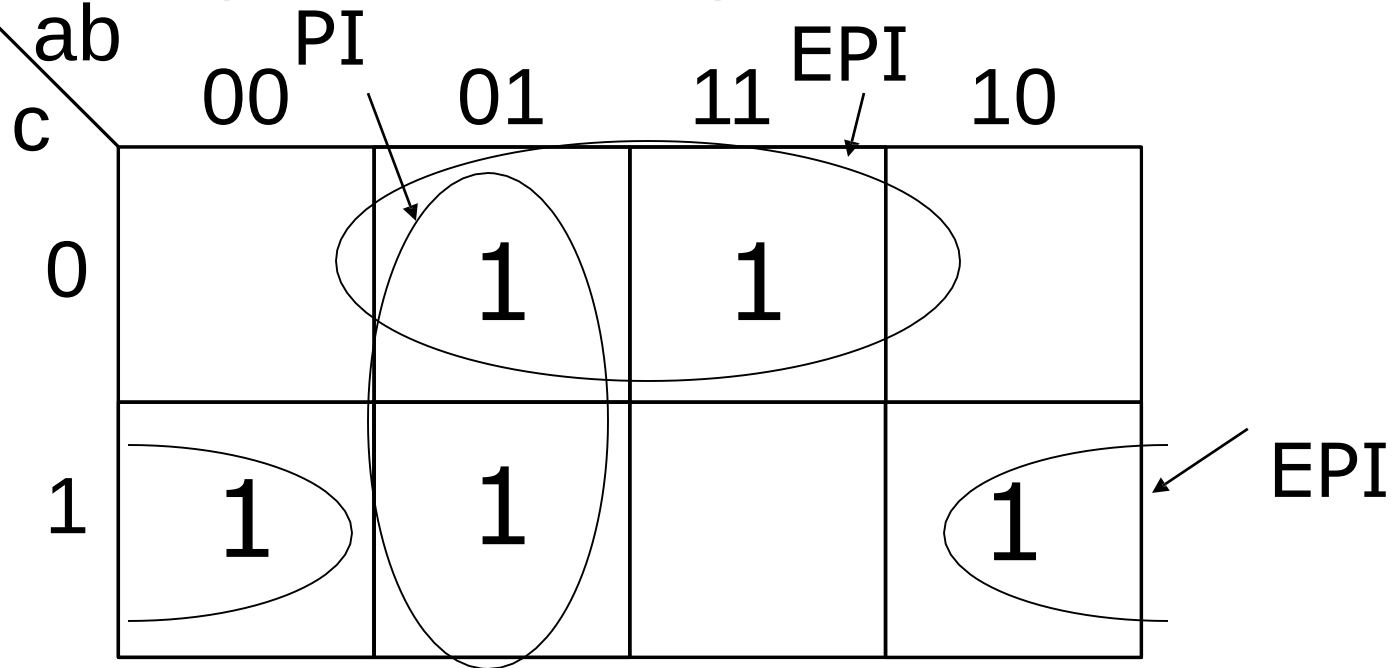
Step 3: Identify Essential Prime Implicants



$$F(a,b,c) = \sum m(1,2,3,5,6)$$

Three-Variable K-Map Example

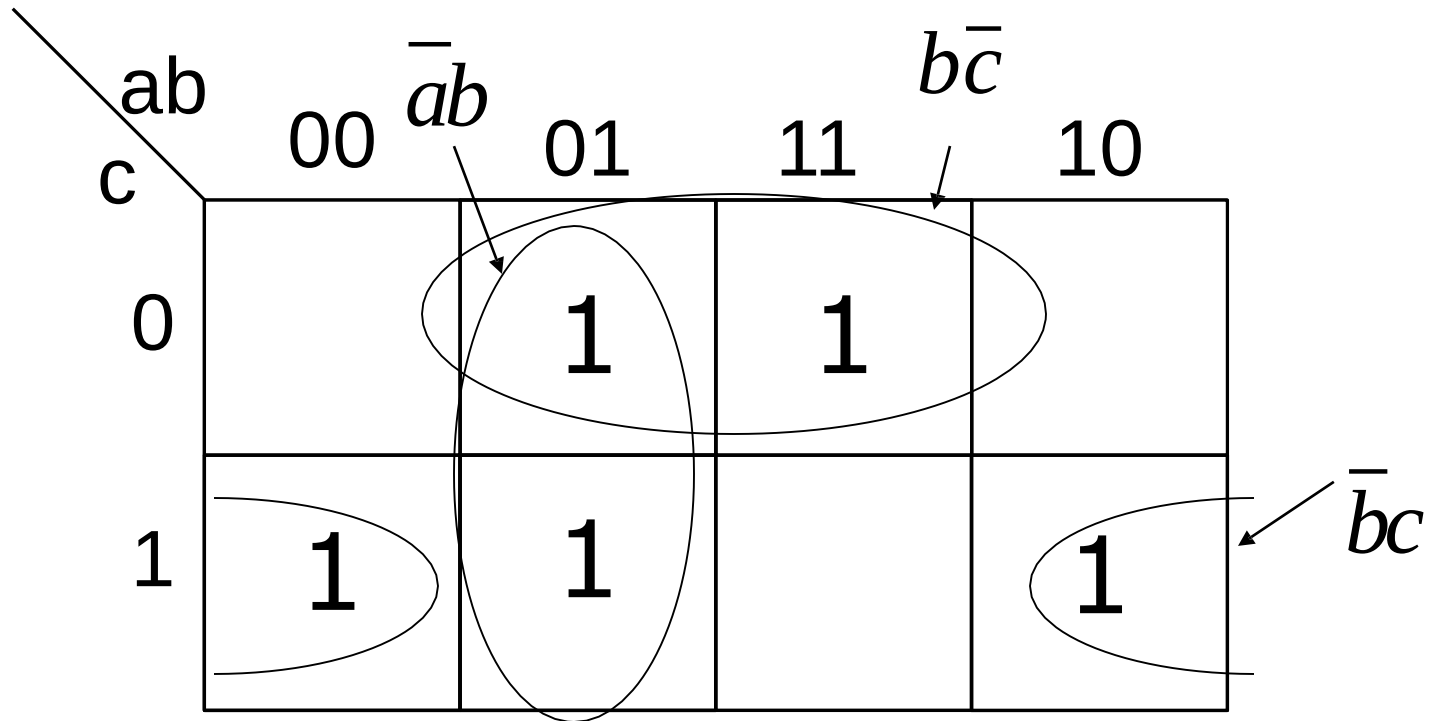
Step 4: Select minimum subset of remaining Prime Implicants to complete the cover.



$$F(a, b, c) = \sum m(1, 2, 3, 5, 6)$$

Three-Variable K-Map Example

Step 5: Read the map.



$$F(a, b, c) = \sum m(1, 2, 3, 5, 6)$$

Solution

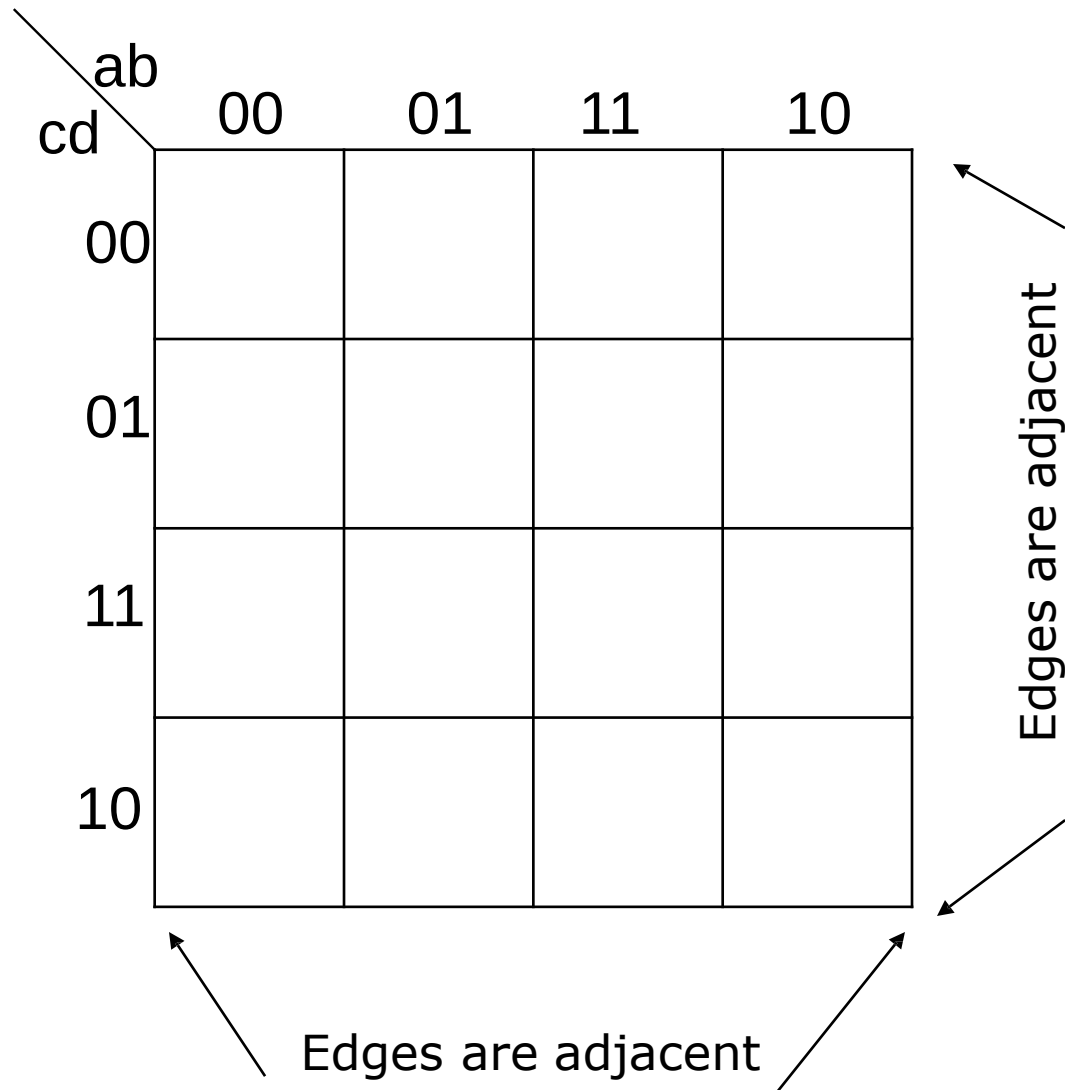
$$F(a, b, c) = \bar{a}b + b\bar{c} + \bar{b}c = \bar{a}b + b \oplus c$$

Four Variable Examples

Four-variable K-Map

		ab			
		00	01	11	10
cd	00	m_0	m_4	m_{12}	m_8
	01	m_1	m_5	m_{13}	m_9
	11	m_3	m_7	m_{15}	m_{11}
	10	m_2	m_6	m_{14}	m_{10}

Four-variable K-Map



Four-Variable K-Maps

		CD			
		00	01	11	10
AB	00	1	0	0	0
	01	0	0	0	0
	11	0	0	0	0
	10	1	0	0	0

$$f = \sum(0,8) = \bar{B} \cdot \bar{C} \cdot \bar{D}$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	0	1	0	0
	11	0	1	0	0
	10	0	0	0	0

$$f = \sum(5,13) = B \cdot \bar{C} \cdot D$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	0	0	0	0
	11	0	1	1	0
	10	0	0	0	0

$$f = \sum(13,15) = A \cdot B \cdot D$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	0	0	1
	11	0	0	0	0
	10	0	0	0	0

$$f = \sum(4,6) = \bar{A} \cdot B \cdot \bar{D}$$

		CD			
		00	01	11	10
AB	00	0	0	1	1
	01	0	0	1	1
	11	0	0	0	0
	10	0	0	0	0

$$f = \sum(2,3,6,7) = \bar{A} \cdot C$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	0	0	1
	11	1	0	0	1
	10	0	0	0	0

$$f = \sum(4,6,12,14) = B \cdot \bar{D}$$

		CD			
		00	01	11	10
AB	00	0	0	1	1
	01	0	0	0	0
	11	0	0	0	0
	10	0	0	1	1

$$f = \sum(2,3,10,11) = \bar{B} \cdot C$$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	0	0	0	0
	11	0	0	0	0
	10	1	0	0	1

$$f = \sum(0,2,8,10) = \bar{B} \cdot \bar{D}$$

Four-Variable K-Maps

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	0	0	0	0
	10	0	0	0	0

$$f = \sum(4,5,6,7) = \bar{A} \bullet B$$

		CD			
		00	01	11	10
AB	00	0	0	1	0
	01	0	0	1	0
	11	0	0	1	0
	10	0	0	1	0

$$f = \sum(3,7,11,15) = C \bullet D$$

		CD			
		00	01	11	10
AB	00	1	0	1	0
	01	0	1	0	1
	11	1	0	1	0
	10	0	1	0	1

$$f = \sum(0,3,5,6,9,10,12,15)$$

$$f = A \otimes B \otimes C \otimes D$$

		CD			
		00	01	11	10
AB	00	0	1	0	1
	01	1	0	1	0
	11	0	1	0	1
	10	1	0	1	0

$$f = \sum(1,2,4,7,8,11,13,14)$$

$$f = A \oplus B \oplus C \oplus D$$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	0	1	1	0
	11	0	1	1	0
	10	0	1	1	0

$$f = \sum(1,3,5,7,9,11,13,15)$$

$$f = D$$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	0	1

$$f = \sum(0,2,4,6,8,10,12,14)$$

$$f = \bar{D}$$

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	1	1	1	1
	10	0	0	0	0

$$f = \sum(4,5,6,7,12,13,14,15)$$

$$f = B$$

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	0	0	0	0
	11	0	0	0	0
	10	1	1	1	1

$$f = \sum(0,1,2,3,8,9,10,11)$$

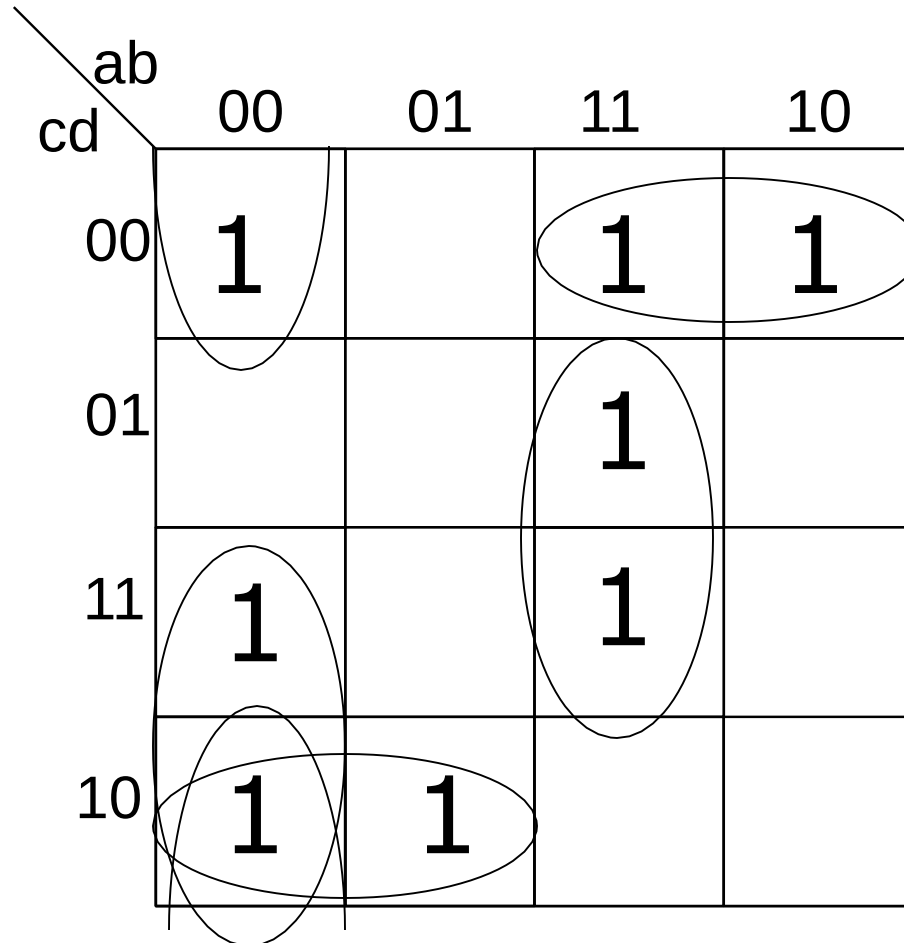
$$f = \bar{B}$$

Example

- Use a K-Map to simplify the following Boolean expression

$$F(a, b, c, d) = \sum m(0, 2, 3, 6, 8, 12, 13, 15)$$

Four-variable K-Map



A 4x4 Karnaugh map for four variables (a, b, c, d). The columns are labeled 'ab' with values 00, 01, 11, 10. The rows are labeled 'cd' with values 00, 01, 11, 10. The map contains 1s in the following cells: (00,00), (11,00), (10,00), (11,01), (11,11), (00,11), (00,10), and (01,10). There are seven groupings of 1s: a horizontal group of three 1s in the top row (00,00), (11,00), and (10,00); a vertical group of two 1s in the third column (11,01) and (11,11); a vertical group of two 1s in the first column (00,11) and (00,10); a horizontal group of two 1s in the bottom row (00,10) and (01,10); a vertical group of two 1s in the third column (11,01) and (11,11); a horizontal group of two 1s in the top row (11,00) and (10,00); and a vertical group of two 1s in the first column (00,11) and (00,10).

cd \ ab	00	01	11	10
00	1		1	1
01			1	
11	1		1	
10	1	1		

$$F = \sum m(0,2,3,6,8,12,13,15)$$

Four-variable K-Map

		ab			
		00	01	11	10
cd	00	1		1	1
	01			1	
	11	1		1	
	10	1	1		

$$F = \overline{a}\overline{b}\overline{c}\overline{d} + \overline{a}\overline{b}c\overline{d} + \overline{a}c\overline{d}\overline{d} + abd + a\overline{c}\overline{d}$$

Four-Variable K-Maps Examples

		CD			
		00	01	11	10
AB	00	1	1		1
	01	1	1		1
	11	1	1		1
	10	1	1		

		CD			
		00	01	11	10
AB	00	1	1		1
	01				1
	11				
	10	1	1		1

		CD			
		00	01	11	10
AB	00				
	01	1	1	1	
	11	1	1		1
	10	1			

		CD			
		00	01	11	10
AB	00		1	1	
	01	1	1	1	1
	11	1		1	1
	10			1	

		CD			
		00	01	11	10
AB	00				
	01				
	11				
	10				

		CD			
		00	01	11	10
AB	00				
	01				
	11				
	10				

Five Variable K-Maps

$$F(a,b,c,d,e)$$

Five variable K-map

Use two four variable K-maps

_____ $A=1$

_____ $A=0$

Use Two Four-variable K-Maps

A=0 map

		bc			
		00	01	11	10
de	00				
	01				
	11				
	10				

A=1 map

		bc			
		00	01	11	10
de	00				
	01				
	11				
	10				

Five variable example

$$F(a, b, c, d, e) = \sum m(5, 7, 13, 15, 21, 23, 29, 31)$$

Use Two Four-variable K-Maps

A=0 map

de \ bc	bc			
	00	01	11	10
00				
01		1	1	
11		1	1	
10				

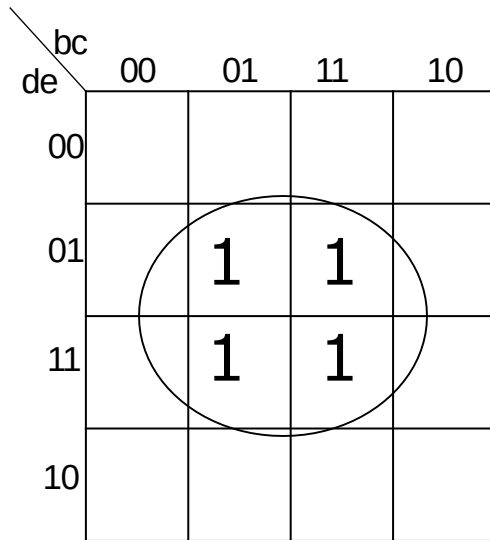
A=1 map

de \ bc	bc			
	00	01	11	10
00				
01		1	1	
11		1	1	
10				

$$F(a,b,c,d,e) = \sum m(5,7,13,15,21,23,29,31)$$

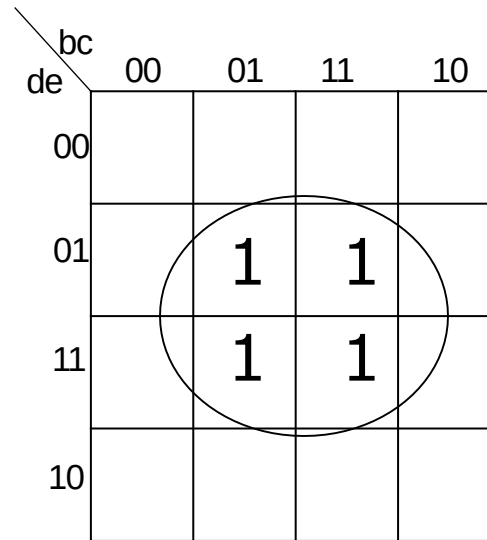
Use Two Four-variable K-Maps

A=0 map



$$F_1 = \bar{d}(a)$$

A=1 map

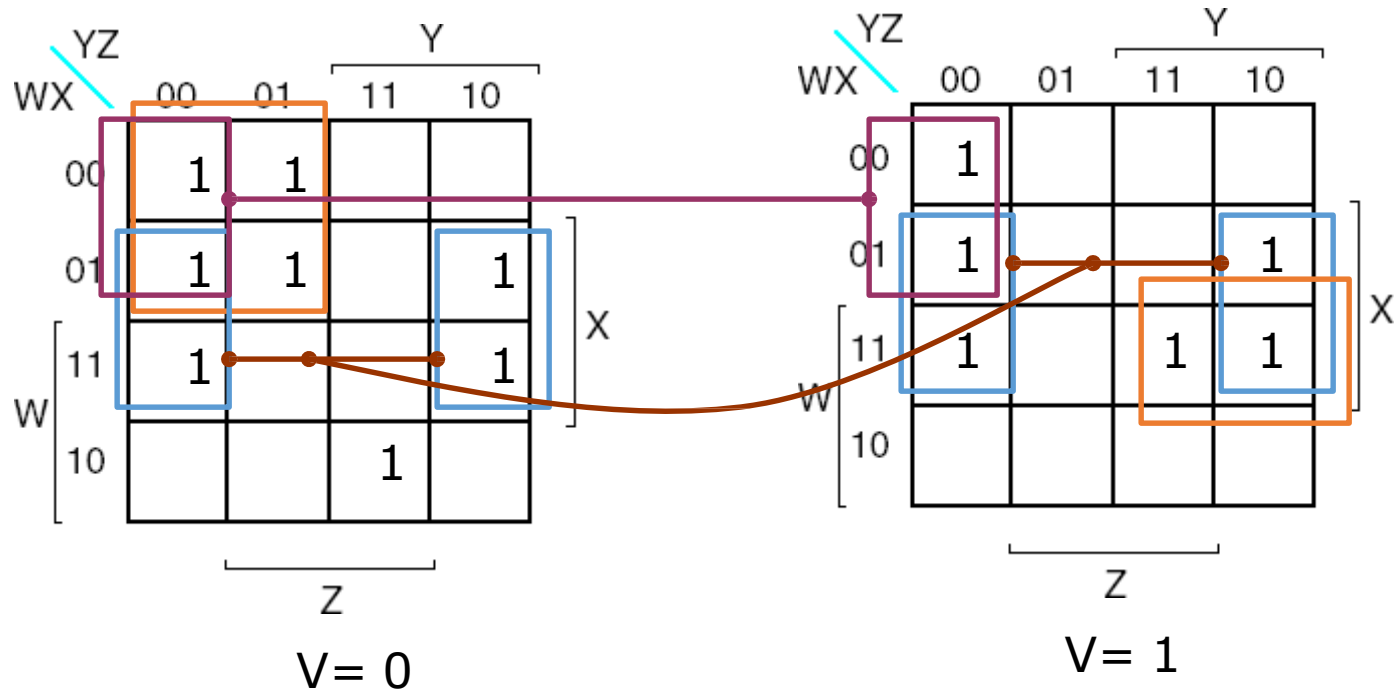


$$F_2 = a(a)$$

Five variable example

$$F = F_1 + F_2 = \bar{a}(ce) + a(ce) = ce$$

Simplify $f(V,W,X,Y,Z)=\Sigma m(0,1,4,5,6,11,12,14,16,20,22,28,30,31)$



$$f = XZ'$$

$$\Sigma m(4,6,12,14,20,22,28,30)$$

$$+ V'W'Y' \quad \Sigma m(0,1,4,5)$$

$$+ W'Y'Z' \quad \Sigma m(0,4,16,20)$$

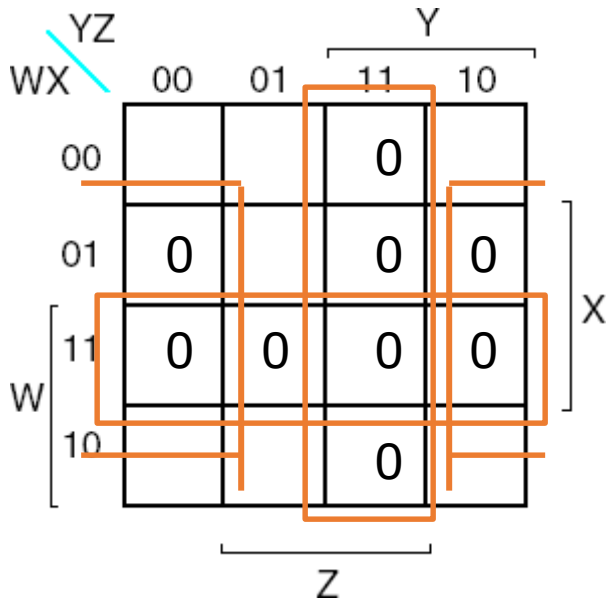
$$+ VWXY \quad \Sigma m(30,31)$$

$$+ V'WX'YZ \quad m11$$

PoS Optimization from SoP

$$F(W,X,Y,Z) = \sum m(0,1,2,5,8,9,10)$$

$$= \prod M(3,4,6,7,11,12,13,14,15)$$



$$F(W,X,Y,Z) = (W' + X')(Y' + Z')(X' + Z)$$

Or,

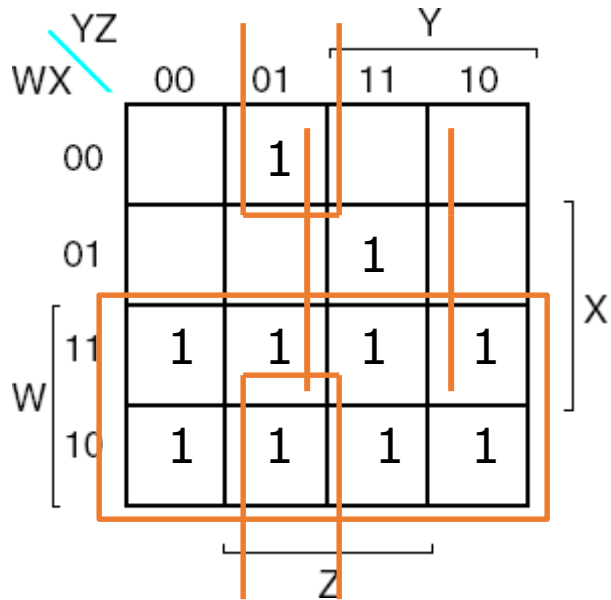
$$F(W,X,Y,Z) = X'Y' + X'Z' + W'Y'Z$$

Which one is the minimal one?

SoP Optimization from PoS

$$F(W,X,Y,Z) = \prod M(0,2,3,4,5,6)$$

$$= \sum m(1,7,8,9,10,11,12,13,14,15)$$



$$F(W,X,Y,Z) = W + XYZ + X'Y'Z$$

Plotting POS Functions

Don't-care Conditions

- In certain problems, some outputs are not specified.
- These outputs can be either '1' or '0'.
- They are called **don't-care conditions**, denoted by X (or sometimes, d).
- Example: An odd parity generator for BCD code which has 6 unused combinations.

- Don't-care conditions can be used to help simplify Boolean expression further in K-maps.
- They could be chosen to be either '1' or '0', depending on which gives the simpler expression.
- We usually use the notation Σd to denote the set of don't-care minterms. For example, the function P in the odd-parity generator for BCD can be written as:

$$P = \Sigma m(0, 3, 5, 6, 9) + \Sigma d(10, 11, 12, 13, 14, 15)$$

Don't-care Conditions

■ For comparison:

❖ WITHOUT don't-cares:

$$P = A'.B'.C'.D' + A'.B'.C.D + A'.B.C'.D \\ + A'.B.C.D' + A.B'.C'.D$$

❖ WITH don't-cares:

$$P = A'.B'.C'.D' + B'.C.D + B.C'.D \\ + B.C.D' + A.D$$

		C			
		D			
AB	CD	00	01	11	10
A	00	1		1	
	01		1		1
	11				
	10		1		

		C			
		D			
AB	CD	00	01	11	10
A	00	1		1	
	01		1		1
	11	X	X	X	X
	10		1	X	X

Four-variable K-Map

		ab			
		00	01	11	10
cd	00	1	d	1	1
	01			1	d
	11	d		1	
	10	1	1		

$$F(a, b, c, d) = \sum m(0, 2, 6, 8, 12, 13, 15) + d(3, 4, 9)$$

Four-variable K-Map

		ab			
		00	01	11	10
cd	00	1	d	1	1
	01			1	d
	11	d		1	
	10	1	1		

$$F = a\bar{c} + \bar{a}\bar{d} + abd$$